

2.7.26

EE25BTECH11014 - Bhoomika Lokesh

Question: Find the area of the quadrilateral $ABCD$, whose vertices are $\mathbf{A}(-3, -1)$, $\mathbf{B}(-2, -4)$, $\mathbf{C}(4, -1)$ and $\mathbf{D}(3, 4)$

Solution: Given the points,

Name	Point
A	$\begin{pmatrix} -3 \\ -1 \end{pmatrix}$
B	$\begin{pmatrix} -2 \\ -4 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$
D	$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

TABLE 0: Variables Used

We can say that,

$$Area(ABCD) = Area(ABC) + Area(ADC) \quad (0.1)$$

Formulae:

$$Area(\triangle ABC) = \frac{1}{2} \|(\mathbf{B} - \mathbf{A}) \times (\mathbf{B} - \mathbf{C})\| \quad (0.2)$$

$$Area(\triangle ADC) = \frac{1}{2} \|(\mathbf{D} - \mathbf{A}) \times (\mathbf{D} - \mathbf{C})\| \quad (0.3)$$

$$(\mathbf{B} - \mathbf{A}) = \begin{pmatrix} -2 \\ -4 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \quad (0.4)$$

$$(\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -2 \\ -4 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ -3 \end{pmatrix} \quad (0.5)$$

$$\|(\mathbf{B} - \mathbf{A}) \times (\mathbf{B} - \mathbf{C})\| = \left\| \begin{pmatrix} 1 \\ -3 \end{pmatrix} \times \begin{pmatrix} -6 \\ -3 \end{pmatrix} \right\| = |-3 - (-6)(-3)| = |-3 - 18| = 21 \quad (0.6)$$

and

$$(\mathbf{D} - \mathbf{A}) = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \end{pmatrix} \quad (0.7)$$

$$(\mathbf{D} - \mathbf{C}) = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \quad (0.8)$$

$$\|(\mathbf{D} - \mathbf{A}) \times (\mathbf{D} - \mathbf{C})\| = \left\| \begin{pmatrix} 6 \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ 5 \end{pmatrix} \right\| = |(6)(5) - 5(-1)| = (30 + 5) = 35 \quad (0.9)$$

$$Area(ABCD) = \frac{1}{2} |(-21)| + \frac{1}{2} |(35)| = \frac{56}{2} = 28 \text{ sq. units} \quad (0.10)$$

$\therefore$ The Area of the quadrilateral $ABCD$ is 28 sq.units.

See Fig. 0.1

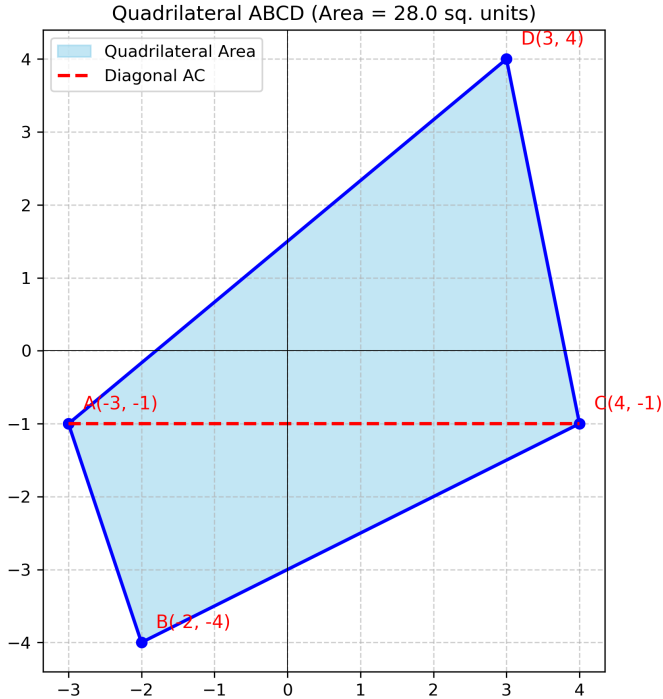


Fig. 0.1: Quadrilateral ABCD